

Circadian and Seasonal Variation in Headache

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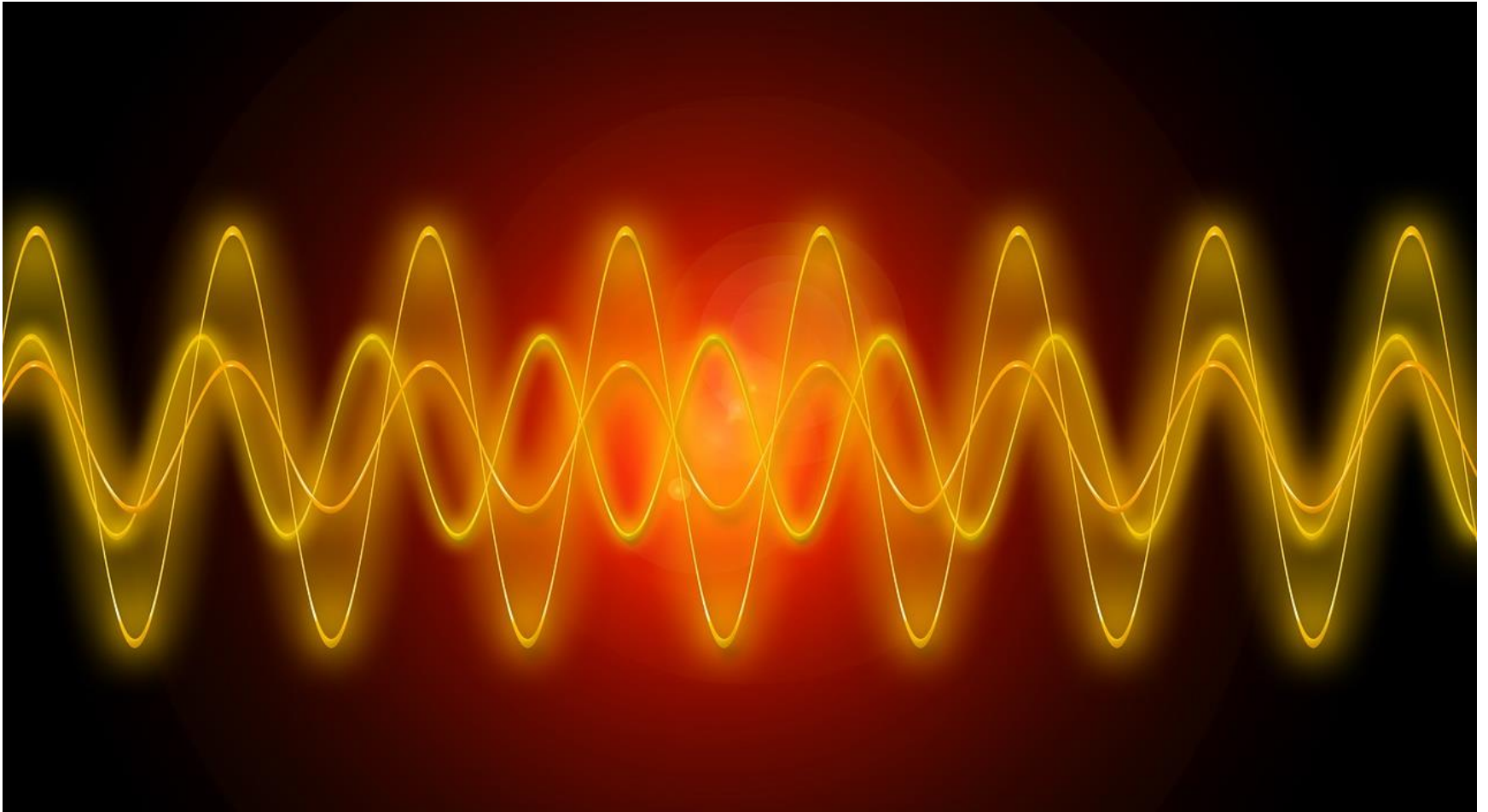


HARVARD
MEDICAL SCHOOL

Disclosures

- GlaxoSmith Kline
 - Statistical/Methodological Consulting
- Nex7, LLC
 - StatReviewer: Co-Founder, Chief Scientist
- Statistical Editor/Consultant
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 - *Headache*
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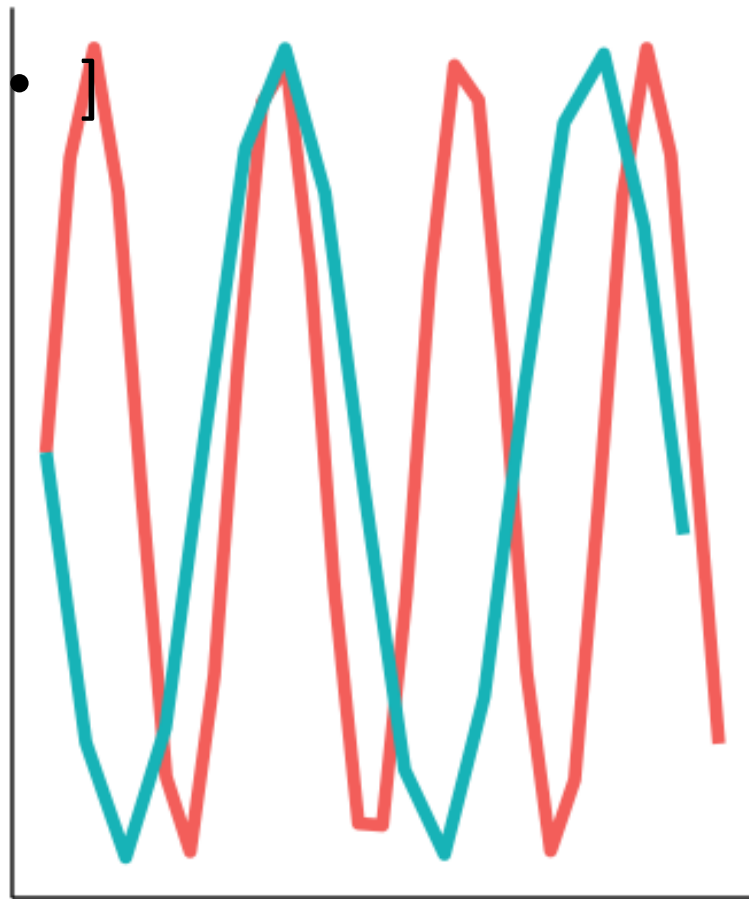
Headache as a Wave



Outline

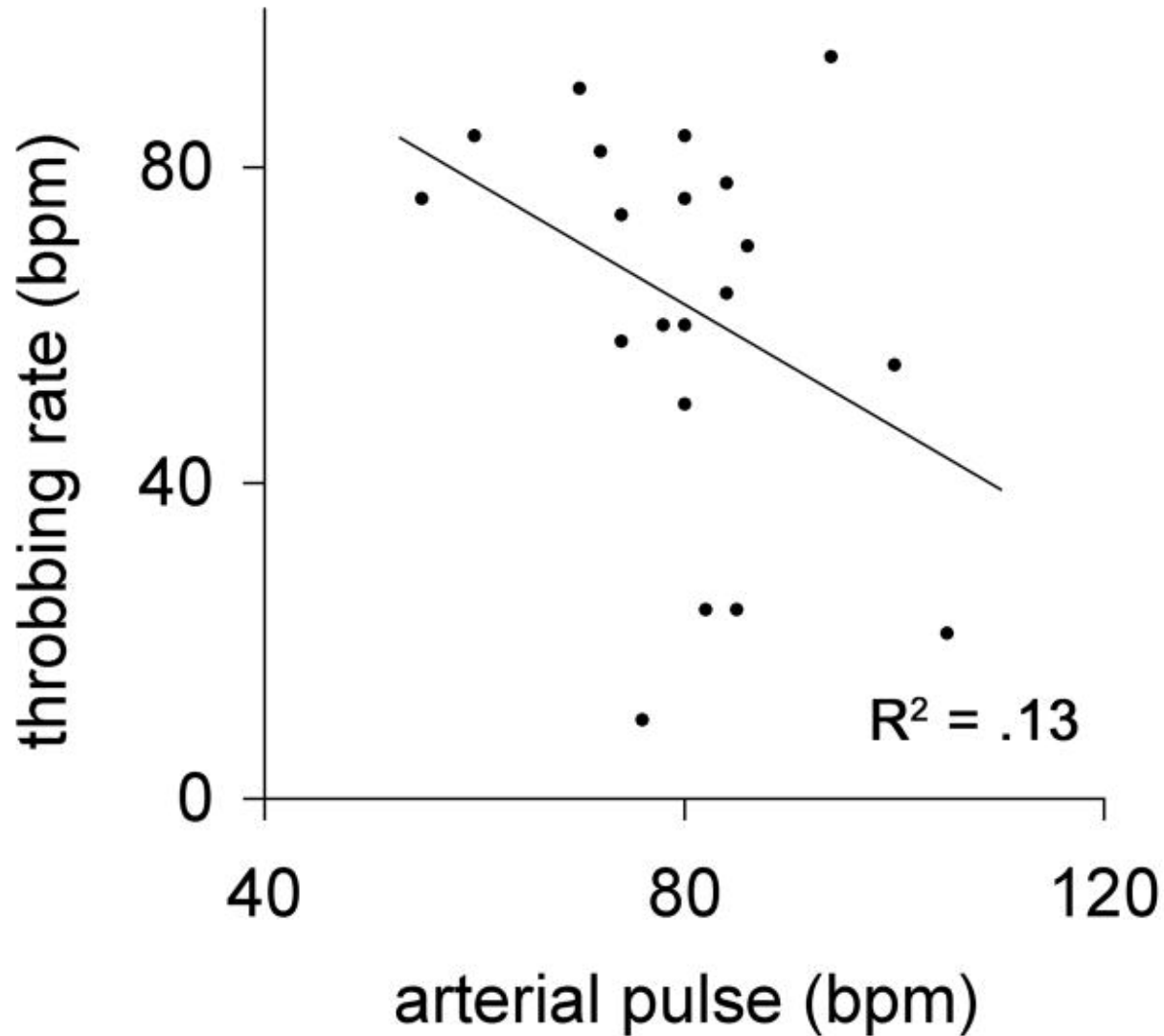
- Cycles in migraine-related throbbing pain perception
- Circadian rhythms
- Weekly attack distributions
- Seasonal attack patterns
- Implications and management

Throbbing Rate and Arterial Pulse Rate



- N = 29 individuals with throbbing dental pain
- Recorded throbbing sensations and arterial pulse rate over time

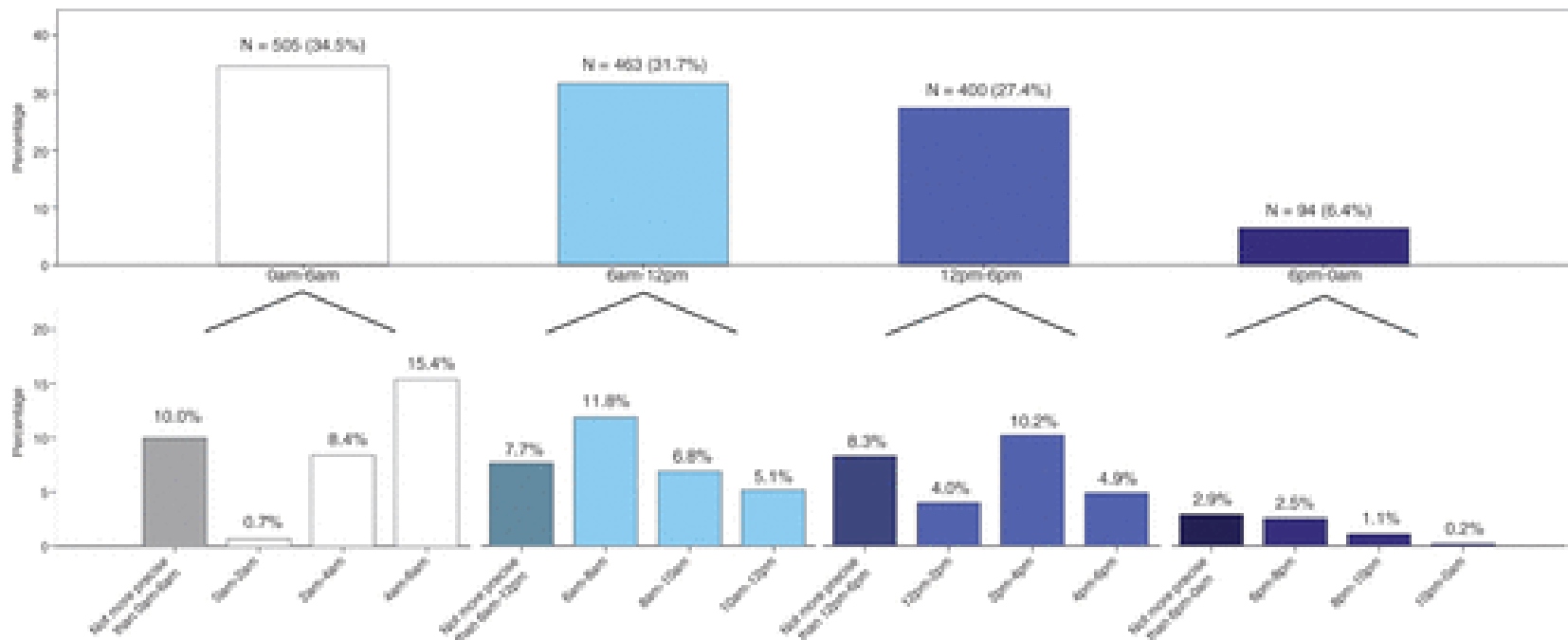
Pulse and Throbbing Rate



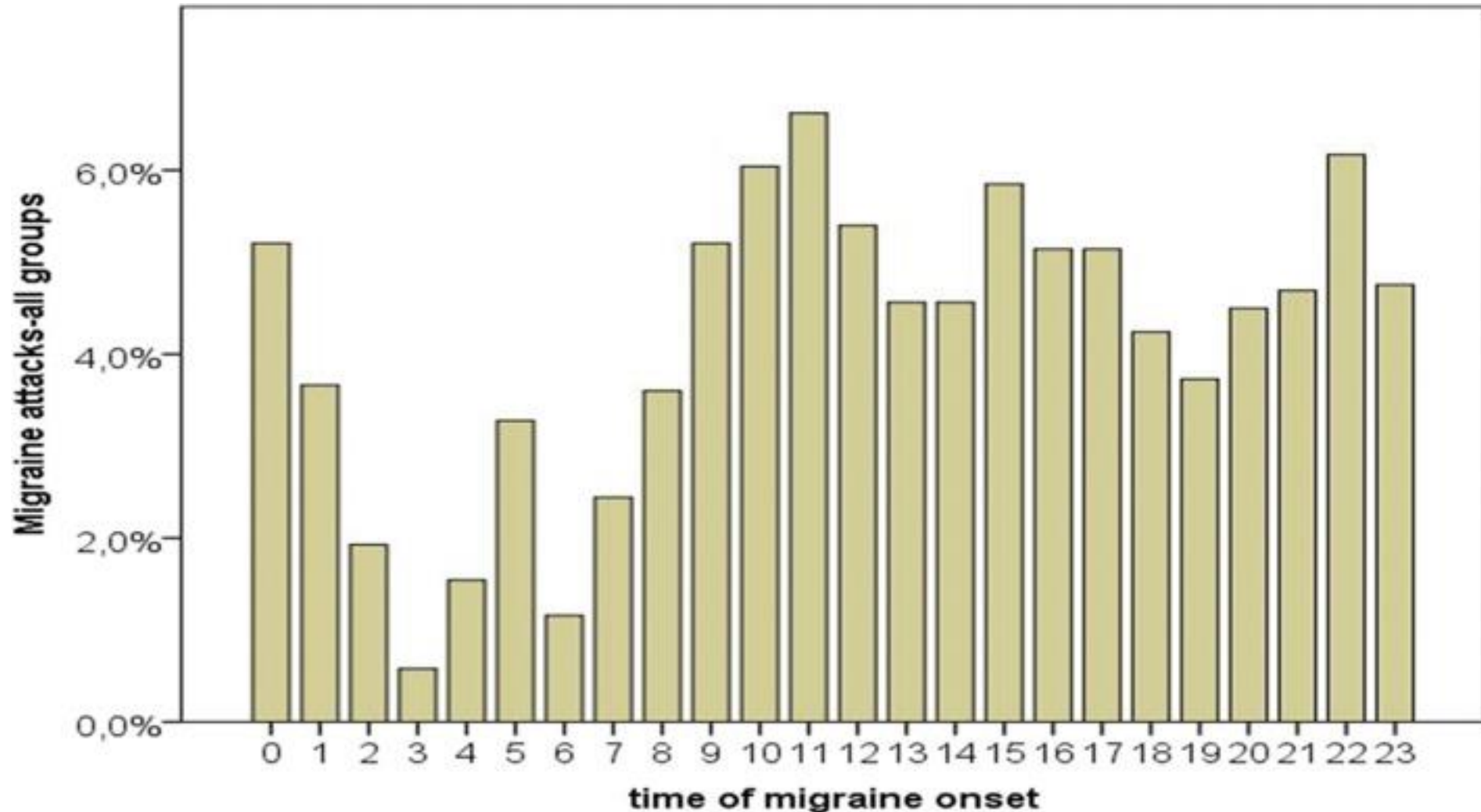
Circadian Cycles in Migraine

- Circadian ‘rhythms’, or 24 hour cycles, are common in living things.
 - Sleep cycles, feeding cycles, body temperature, hormone production
- Migraine attacks have long been thought to have circadian periods, but evidence is difficult to establish
 - Cross-sectional vs. daily diary
 - Differential reliability (night vs. day)
- Evidence to support the hypothesis is sought by examining the migraine onset time

N = 2389 individuals with migraine, cross-sectional survey



N = 786 individuals with migraine,
16,578 daily-diary attacks



24-Hour Distribution of Migraine Attacks

Karl Alstadhaug, MD; Rolf Salvesen, MD, PhD; Svein Bekkelund, MD, PhD

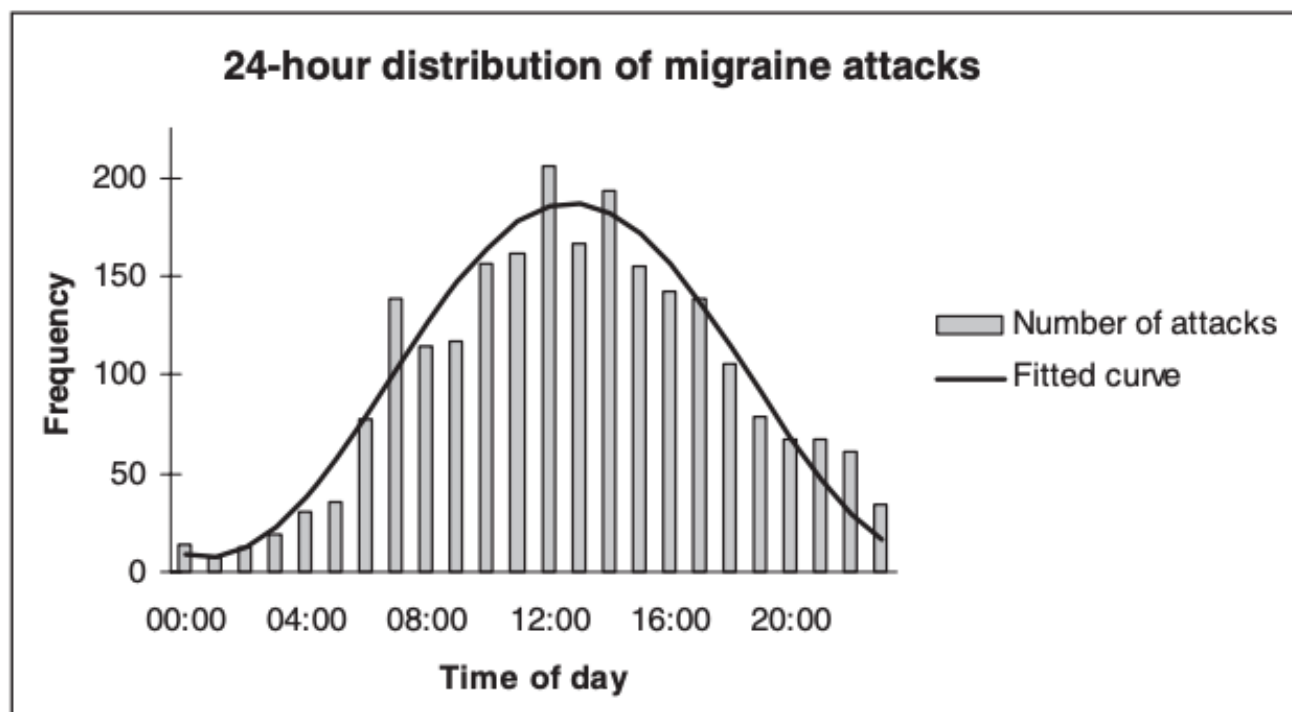
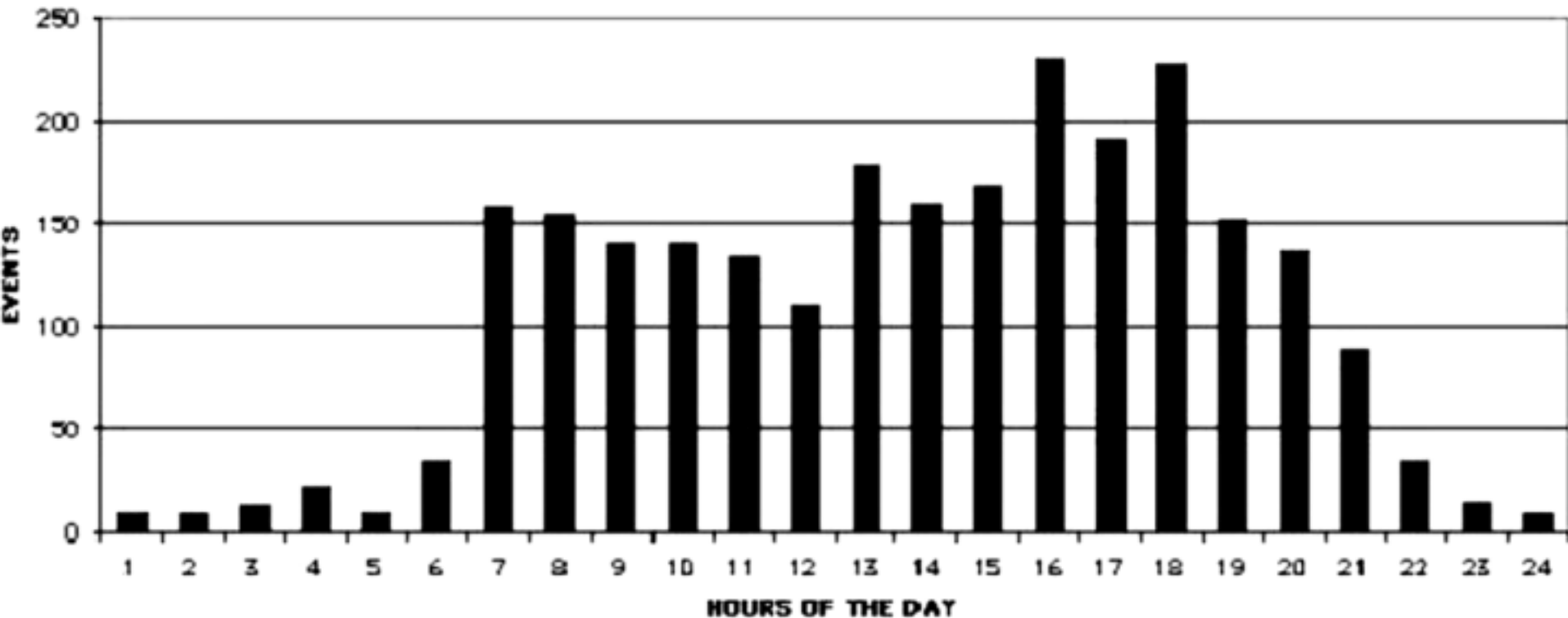


Fig 1.—The 24-hour temporal distribution of migraine is shown. There is a peak of occurrence just after noon. By fitting a sine curve to the frequency histogram a harmonic trend is indicated.

N = 115 Children
2517 daily-diary attacks



Longer Cycles: Weekly, Monthly, Seasonality

- Difficult to study using cross-sectional measures
 - Often requires memory or aggregation of previous attack characteristics
 - When do you experience a “typical attack”?
 - Can rely on population characteristics (e.g., ED visits for migraine)
- Must assess variability in attack patterns
 - Assessed using sampling rates appropriate for the cycle under study
 - Would require long observation periods for within-person variability (e.g., months for weekly patterns, years for monthly)

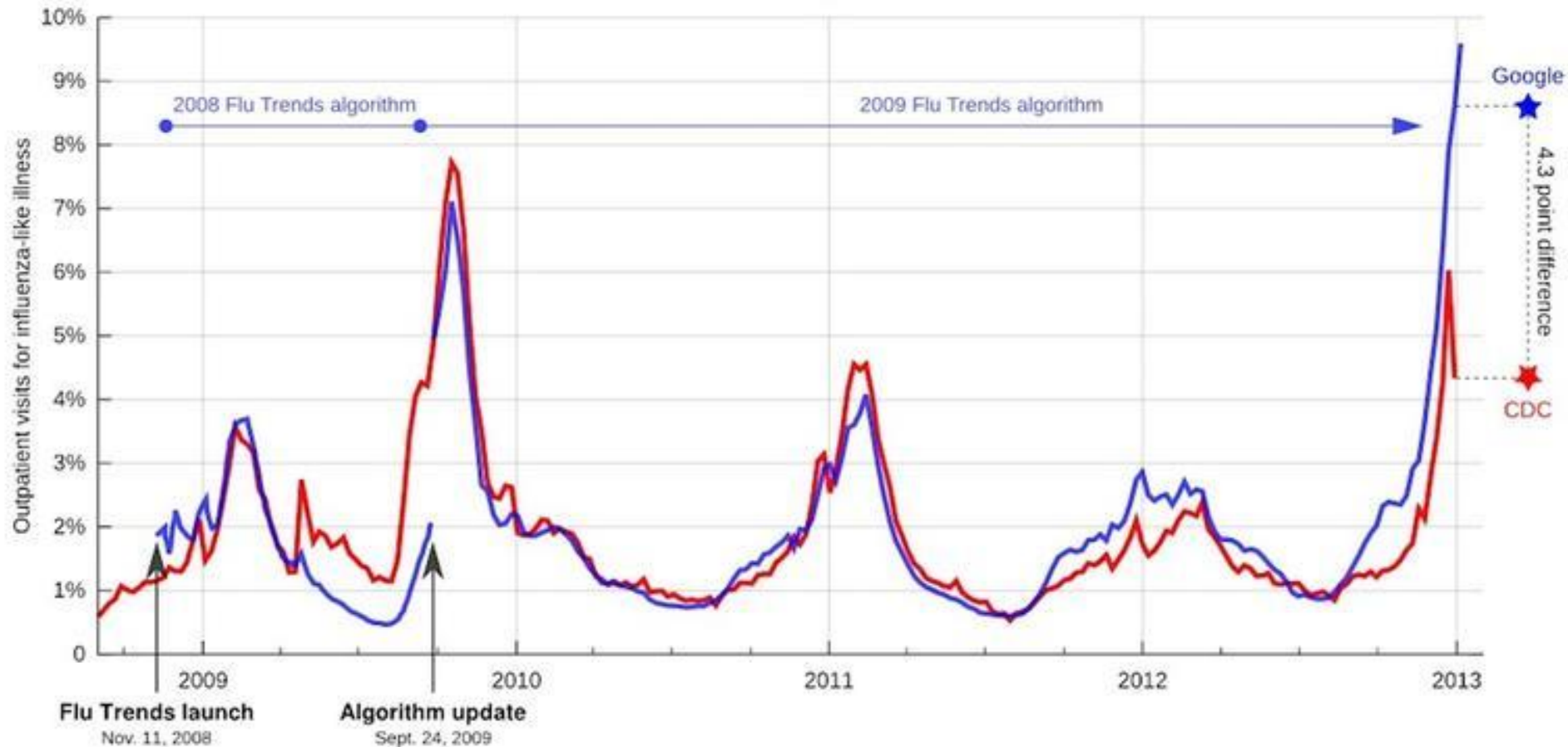
Using Google Searches As Population Interest in Migraine

- Google search represents an interest or 'intention'
 - Treatment
 - Symptoms
 - Triggers/causes
- May not reflect real-time intention (i.e., lagged association with attacks)
 - First experience an attack, then investigate additional treatments or causes

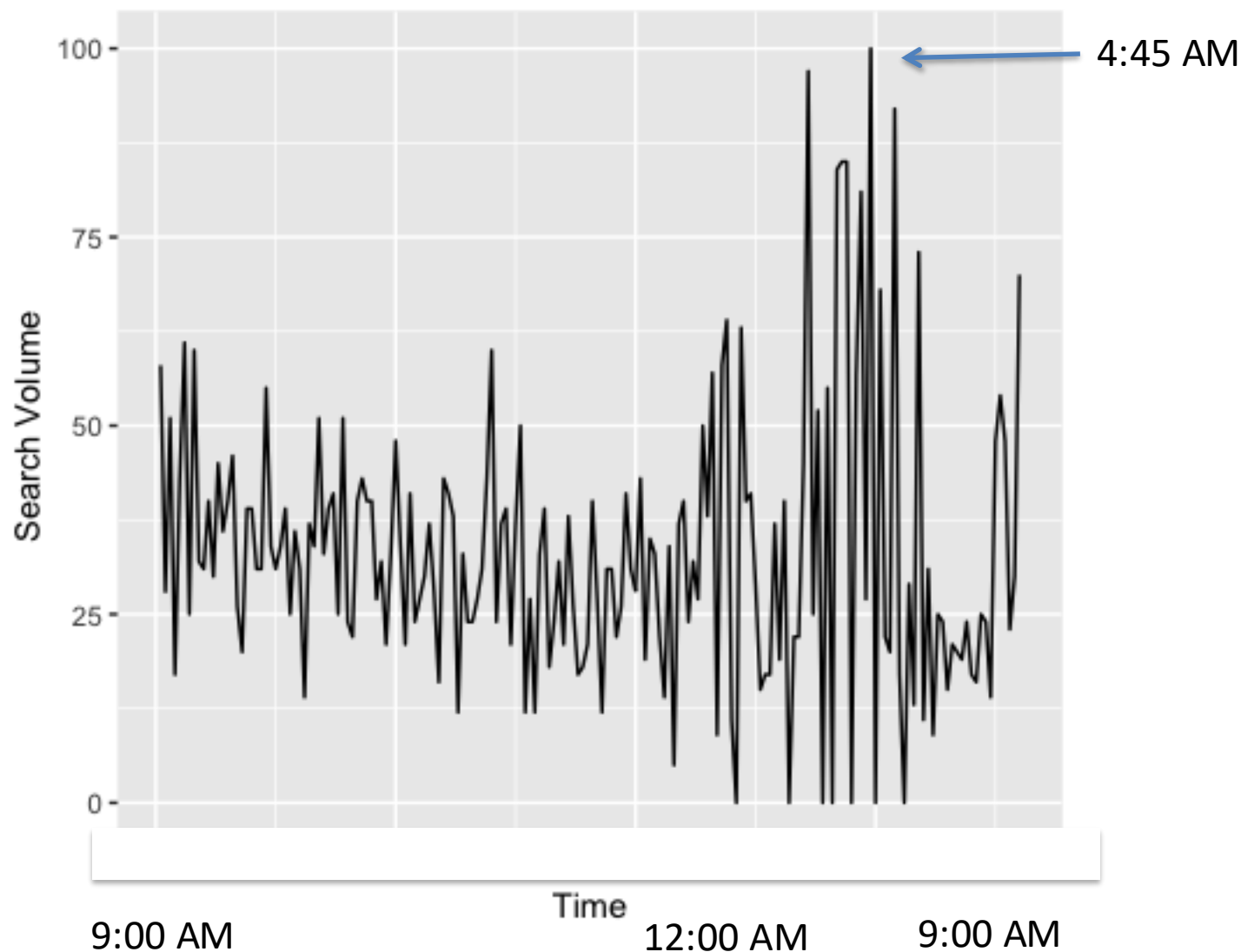


Google Flu Trends

Google Flu Trends U.S. may have diverged again from the CDC data it predicts, but too early to be sure.



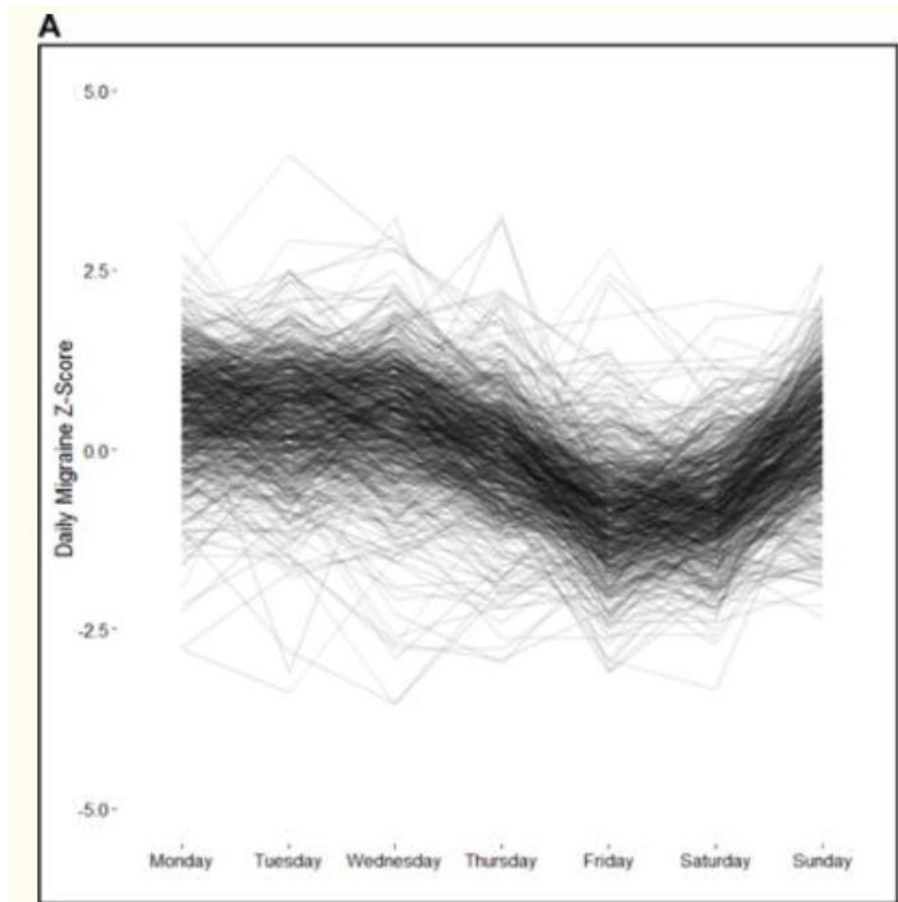
Hourly Google Searches for Migraine: Pennsylvania-today



Using Search Engines to Investigate Shared Migraine Experiences

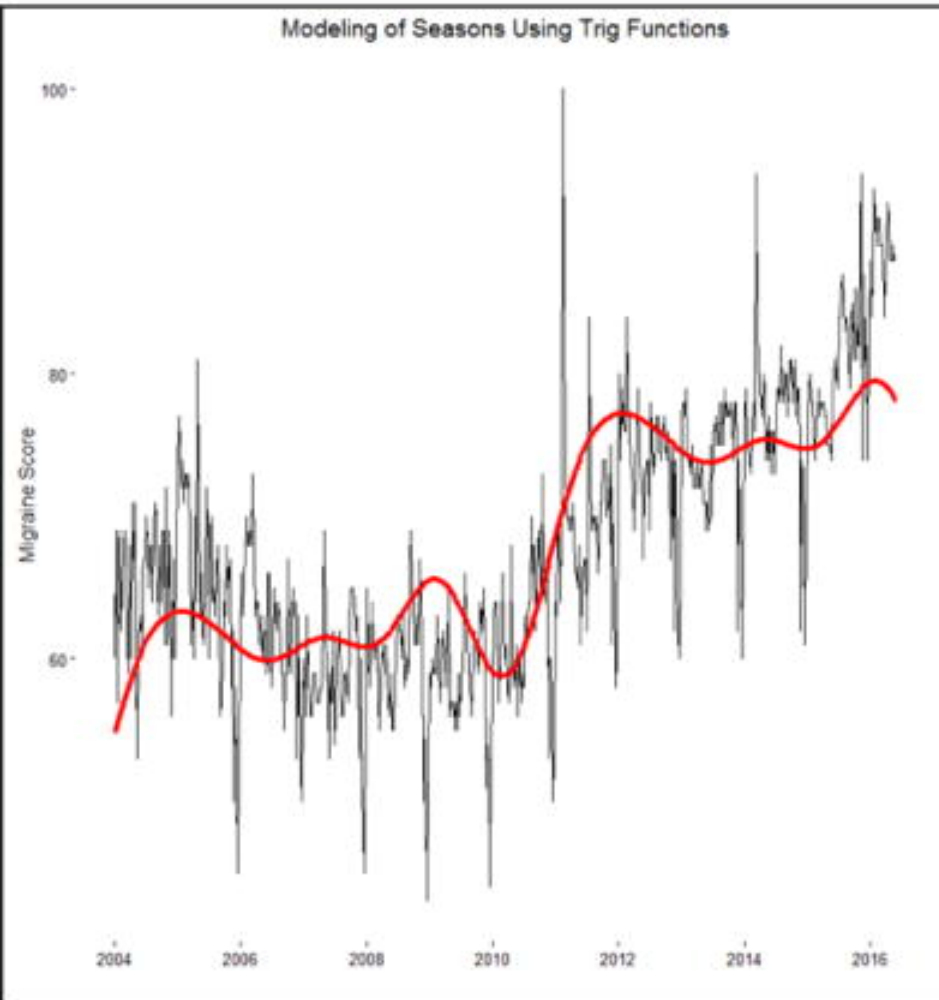
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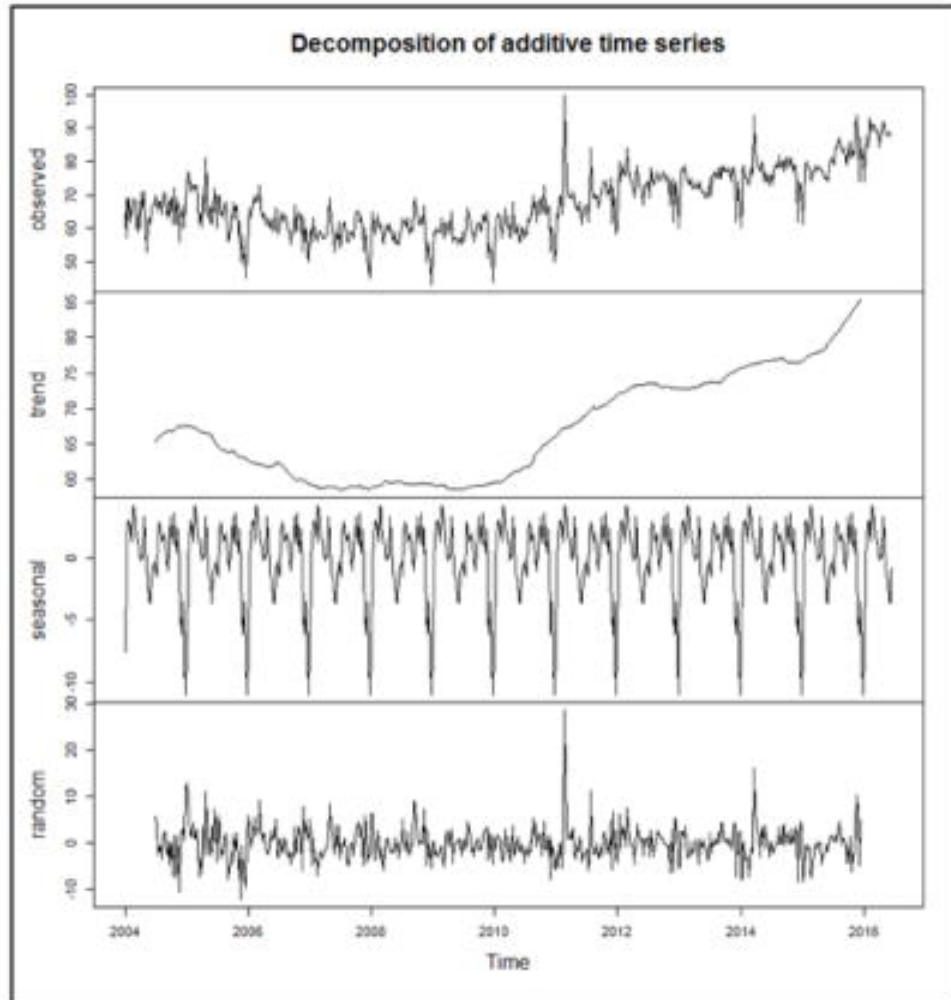


Seasonal Attack Pattern

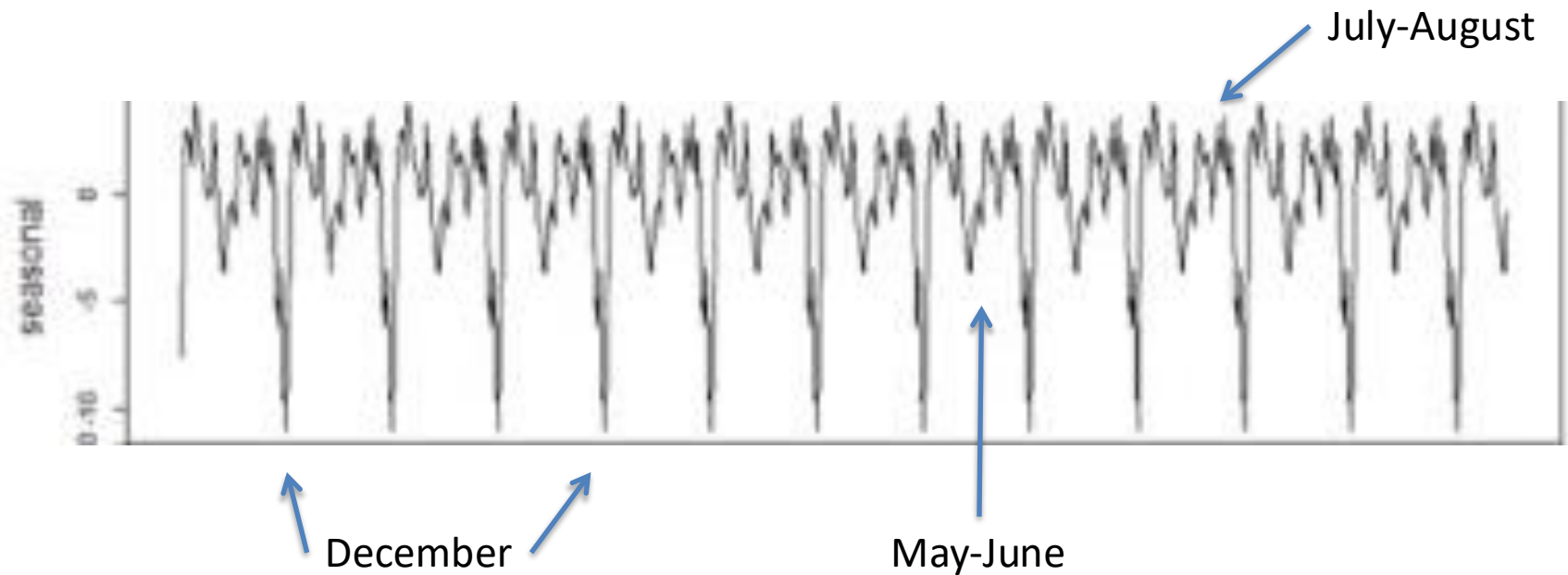
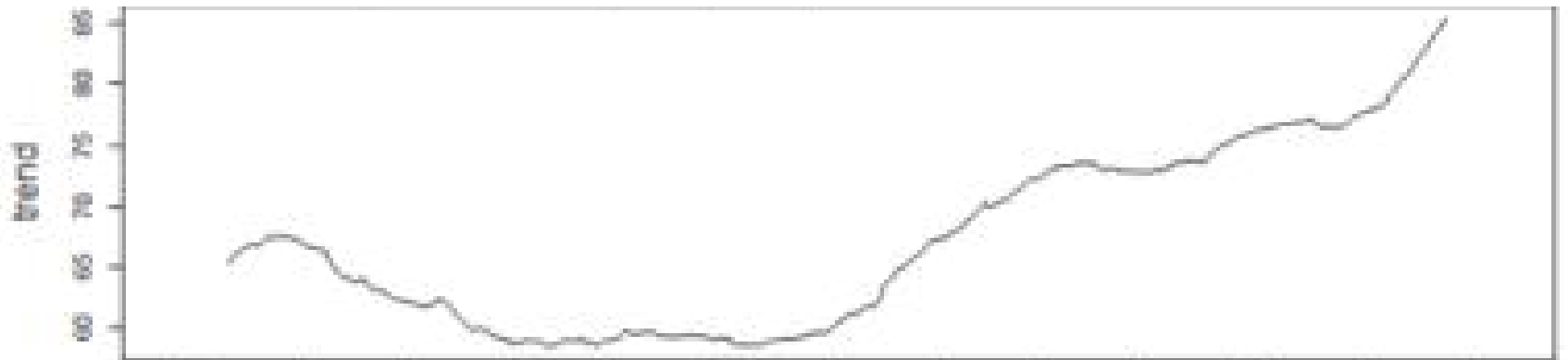
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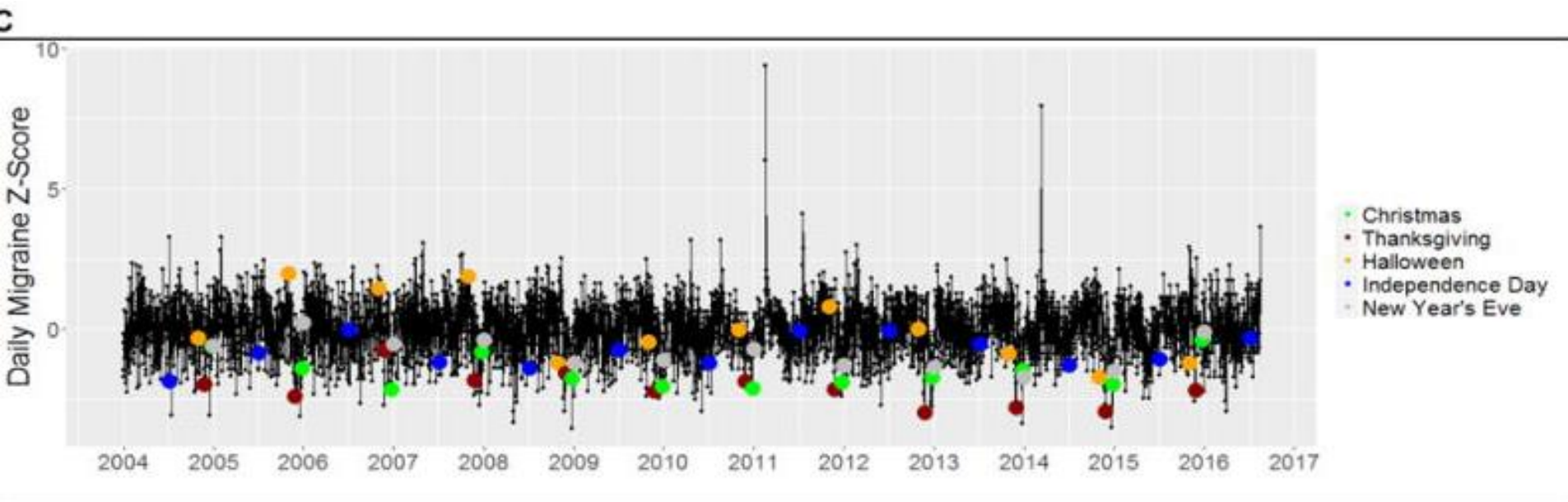
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Seasonal Decomposition



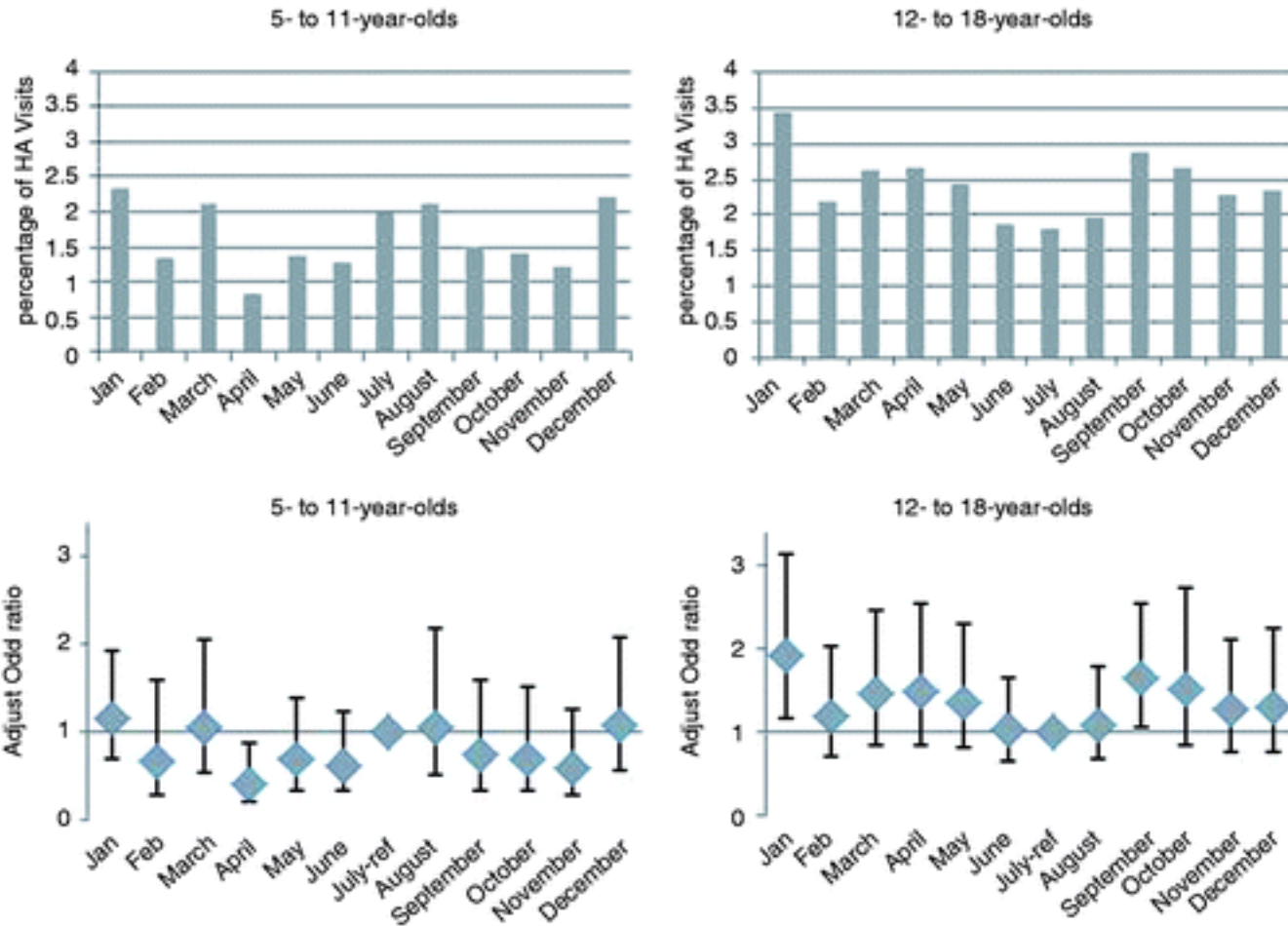
Specific Days



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	Date	Migraine Score	Z-Score	Possible Explanation
Highs	2/17/2011	100	9.39	CBS reporter Serene Branson suffered migraine on national television
	3/11/2014	100	7.96	FDA approves device to treat migraine headaches (Tuesday)
	2/18/2011	76	6.02	Aftermath of CBS reporter Serene Branson suffered migraine on national television
	7/19/2011	100	4.11	Michele Bachmann (Presidential candidate) comes out as a public migraine sufferer (Tuesday)
	8/15/2016	100	3.65	Seattle Times Article on Migraines and Gluten (Monday)
Lows	12/24/2008	32	-3.55	Christmas Eve
	12/24/2014	53	-3.53	Christmas Eve
	12/24/2013	51	-3.38	Christmas Eve
	5/3/2008	46	-3.34	Storms and tornados in central USA (Saturday)
	12/16/2005	28	-3.10	Winter ice storms caused power outages for 3/4 million people along Atlantic coast (Friday)

Seasonal Variation in Emergency Department Visits



Implications of Cyclicity

- Implications for migraine pathophysiology
 - Pulsation frequency implies unique mechanism oscillating at unique frequency
- Aids in identification of associated (causal?) influences
 - Stress (Hourly, Weekly, Monthly)
 - Food patterns
- Focus on risk 'cycles' not just risk periods

Management of Risk Cycles

- Implies periods of both:
 - Low risk (end of week, Nov-Dec)
 - High risk (morning, mid-day, early week, July-Aug)
- Pharmacological strategies might differ if at high risk part of cycle
 - Greater expected headache frequency calls for reconsideration of medication use targets?
- Differential behavioral or logistical management strategies corresponding to co-varying cycle components
 - Barometer
 - School schedule

Assessment of Cycles Within an Individual

- Most data express population cycles
 - Not necessarily the same for an individual
- Assessment strategy must sample variation in headaches and triggers (or influences) according to the nature of the cycle
 - Weekly cycles require daily monitoring
 - Monthly cycles require weekly monitoring
- Great tools now available to assist individuals in monitoring a variety of influences
 - Curelator

Thank You!

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